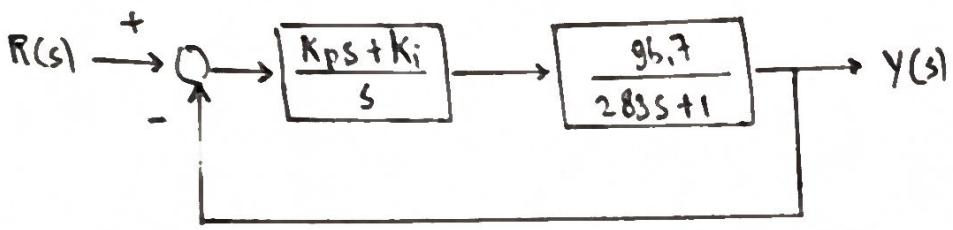
- **Finding K_p and K_i**



Unity feedback because DS18B20 is really fast compared to the plant.

$$CLTF = \frac{\frac{K_p s + K_i}{s}\ \frac{96.7}{2835+1}}{1 + \frac{K_p s + K_i}{s}\ \frac{96.7}{2835+1}} = \frac{96.7K_p s + 96.7K_i}{2835^2 + (1 + 96.7K_p)s + 96.7K_i} = \frac{0.342K_p s + 0.342K_i}{s^2 + (0.0035 + 0.342K_p)s + 0.342K_i}$$

Apply RH.

s^2	1	$0.342K_i$
s^1	$0.0035 + 0.342K_p$	0
s^0	$0.342K_i$	0

$0.0035 + 0.342K_p > 0$

$K_p > -0.01$

$0.342K_i > 0$

$K_i > 0$

Absolute stability ranges

$\boxed{K_p > -0.01 \text{ & } K_i > 0}$

Goals of thermobox : 1. minimal settling time T_s

2. minimal overshoot %OS

3. disturbance rejection

* $T_s = 10$ minutes $= 600$ sec & $\%OS \leq 2\%$.

$$\psi = \sqrt{\frac{\ln^2\left(\frac{\%OS}{100}\right)}{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}} = 0.7797$$

Choose $\psi = 0.8 > 0.7797$

$$T_s = \frac{4}{\psi\omega_n} \Rightarrow \psi\omega_n = \frac{4}{T_s} = \frac{4}{600} = 6.667 \times 10^{-3}\ rads^{-1}$$

$$\omega_n = \frac{6.667 \times 10^{-3}}{0.8} = 8.334 \times 10^{-3}\ rads^{-1}$$

$$2\psi\omega_n = 2 \times 6.667 \times 10^{-3} = 13.334 \times 10^{-3}\ rads^{-1}$$

Second-order prototype : $\frac{\omega_n^2}{s^2 + 2\psi\omega_n s + \omega_n^2}$

$$\therefore K_p = \frac{13.334 \times 10^{-3} - 0.0035}{0.342} = 0.028754 \quad \& \quad K_i = \frac{(8.334 \times 10^{-3})^2}{0.342} = 0.000203$$

$\boxed{K_p = 0.028754 \text{ and } K_i = 0.000203}$